1. Cryptographic assumption attack corpus

Prioritized formal-audit plan · 2026-11-01

1.1. Outcome

This search converts recurring ECC and proof-system slogans into exact counterexamples, conditional repair theorems, model boundaries, or explicit interface gaps. Five high-leverage finite targets are frozen now. Their Lean files do not yet exist, so they are **undischarged pointers**, not verified proofs, vulnerabilities, or security conclusions.

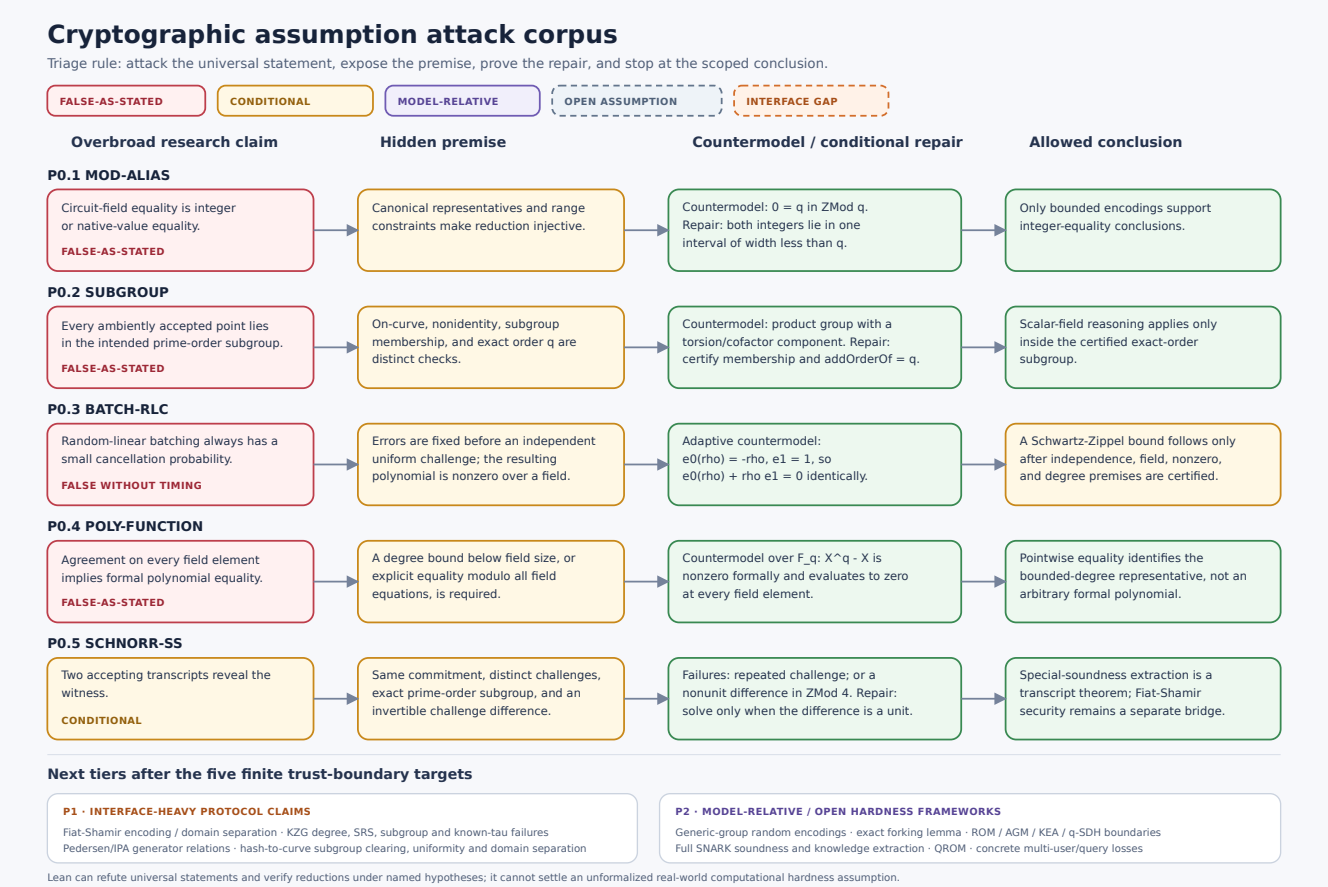


Figure 1: Claim-to-premise triage. Red marks false universal wording; amber marks required hypotheses; green marks the repair that is actually available. Model-relative and open-hardness layers remain separately named.

1.2. Classification rule

Status	Meaning
false-as-stated	A concrete model satisfies the premises and falsifies the conclusion.
conditionally proved	The conclusion follows only after the theorem type exposes every algebraic, order, distributional, encoding, or timing premise.
model-relative	The theorem lives in a named GGM, ROM, AGM, or QROM model; the qualifier cannot be erased.
open computational assumption	Lean may verify reductions and finite controls, but does not settle the real-world unbounded hardness assumption.
unconnected interface	The abstract theorem and implementation endpoint exist, but no checked refinement connects them.

1.3. Frozen first tranche

Target	Countermodel	Conditional repair	Boundary
MOD-ALIAS	`0` and `q` are distinct naturals but equal in `ZMod q`.	Casting is injective for representatives strictly below positive `q`.	A circuit claim still needs a range/encoding refinement.
SUBGROUP	`ZMod 2 × ZMod 3` contains ambient elements outside the designated order-three factor.	Supply exact-order and membership witnesses; require nonidentity when the protocol does.	This is not yet an elliptic-curve decoder or invalid-curve theorem.
BATCH-RLC	After seeing `rho`, choose `e0=-rho`, `e1=1`; the combination vanishes identically.	Fix a nonzero error polynomial before an independent uniform field challenge and apply its degree/root bound.	Fiat–Shamir transcript binding is a separate theorem.
POLY-FUNCTION	`X^q-X` is formally nonzero and zero as a function on `F_q`.	A univariate polynomial of degree below `|F|` that vanishes everywhere is zero.	Commitment degree, SRS, quotient, and evaluation-domain interfaces remain.
SCHNORR-SS	Equal challenges reveal nothing; a nonzero difference in `ZMod 4` may be a nonunit.	Same commitment, distinct challenges, exact prime-order subgroup, and injective scalar encoding give extraction.	Special soundness is not Fiat–Shamir, ROM/QROM, or proof-of-knowledge security.

The five task specifications live under `formal/targets/attack-*.yaml`. A green theorem build will still require independent semantic-fidelity review.

1.4. What to pursue next

Priority 0 composition controls. Cofactor projection and pushforward; constant-hash Fiat–Shamir; Pedersen/IPA relation kernels; zero knowledge versus soundness and proof of knowledge; correlated repetition; correct DLP, CDH, and DDH reduction directions.

Priority 1 protocol interfaces. Exceptional EC formulas; invalid-curve and twist decoding; injective transcript serialization and domain separation; KZG known-`tau`, degree, subgroup, and batching boundaries; PLONK lookup and permutation denominators; circuit-to-native EC refinement; subgroup-scoped random self-reducibility; Pollard-rho random-map assumptions; RFC 9380 suite, uniformity, and cofactor contracts; concrete query and multi-user losses.

Priority 2 foundations. A finite game/oracle framework; exact forking loss; generic-group random encodings; ROM, AGM, KEA, q-SDH, and QROM boundaries; separate SNARK security definitions; then full KZG, Groth16, and PLONK.

1.5. Ninety-day sequence

1. **Days 0–30:** discharge the five finite targets with countermodel and repair in the same module; axiom-audit them and review statement fidelity.
2. **Days 31–60:** formalize cofactor, Fiat–Shamir zero-hash, Pedersen, ZK/soundness, repetition, and KZG toxic-waste algebraic controls.
3. **Days 61–90:** implement a minimal finite transcript/oracle semantics, then prove one batching game and one two-fork extraction game before attempting a generic-group or commitment-security result.

1.6. Source and scope boundary

Primary boundaries were checked against JKSST, Shoup, Pointcheval–Stern, Bellare–Garay–Rabin, KZG, SEC 1, RFC 9380, the AGM paper, Groth16, and PLONK. Repository inputs include `docs/reports/xedni-sadek-20261101/`, `problems/xedni-sadek-explicit-cutoff.md`, `H-XEDN-292264`, and `KN-LIT-c9e59d` on `GAG_q`. Existing STARK semantic reviews already record the modular-alias warning and the `X^2-X` finite-field example; this corpus targets their Lean formalization, not novelty. Full provenance and theorem-shaped statements are in `README.md` beside this report.

Lean can refute universal mathematical statements and verify reductions under named hypotheses. It cannot prove or disprove an unformalized real-world DLP, CDH, q-SDH, KEA, or knowledge-soundness assumption, and it cannot establish that deployed code matches an abstract model without a refinement proof.